

* Choose The Right Answer From The Given Options.[1 Marks Each]

[23]

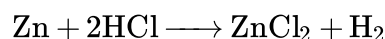
- If 3.01×10^{20} molecules are removed from 98 mg of H_2SO_4 , then number of moles of H_2SO_4 left are:
(A) $0.5 \times 10^{-3}\text{mol}$. (B) $0.1 \times 10^{-3}\text{mol}$.
(C) $9.95 \times 10^{-3}\text{mol}$. (D) $1.66 \times 10^{-3}\text{mol}$.
- 1.020g of metallic oxide contains 0.540g of the metal. If the specific heat of the metal, M is $0.216 \text{ cal deg}^{-1}\text{g}^{-1}$. The molecular formula of its oxide is:
(A) MO (B) M_2O_3 (C) M_2O_4 (D) M_2O
- Total number of orbitals associated with third shell will be _____.
(A) 2. (B) 4. (C) 9. (D) 3.
- The radius of second Bohr's orbits for hydrogen atoms is:
 $h = 6.6262 \times 10^{-34} \text{Js}$, $m_e = 9.109 \times 10^{-31} \text{kg}$, $e_{\text{charge}} = 1.6021 \times 10^{19} \text{C}$
(A) $1.65\text{\AA}$ (B) $4.76\text{\AA}$ (C) $0.529\text{\AA}$ (D) $2.12\text{\AA}$
- The correct set of four quantum numbers for the valence electrons of rubidium atom ($Z = 37$) is:
(A) $5, 0, 0 + \frac{1}{2}$ (B) $5, 1, 0 + \frac{1}{2}$ (C) $5, 1, 1 + \frac{1}{2}$ (D) $5, 0, 1 + \frac{1}{2}$
- Outer electronic configuration of f-block elements is:
(A) $(n + 1)f^{1-14}(n - 1)d^{0-1}ns^2$ (B) $(n - 2)f^{1-14}(n + 1)d^{0-1}ns^2$
(C) $(n - 2)f^{1-14}(n - 1)d^{0-1}ns^2$ (D) None of the above.
- As we go down in the electro-chemical series of metals, the reactivity _____.
(A) Decreases and then increases (B) Increases and then decreases
(C) Decreases (D) Increases
- Which of the following statements is incorrect?
(A) Mendeleev's arranged elements in horizontal rows and vertical columns.
(B) Mendeleev's arranged elements in order of their increasing atomic number.
(C) Mendeleev's system of classifying elements was more elaborate than that of Lother Meyer.
(D) None of the above.
- Which of the following options represents the correct bond order?
(A) $\text{O}_2 > \text{O}_2 > \text{O}_2^+$ (B) $\text{O}_2^- > \text{O}_2 > \text{O}_2^+$

- (C) $O_2 > O_2 < O_2$ (D) $O_2^- < O_2 > O_2^+$
10. H – O – H bond angle in water is:
 (A) 104.5° (B) 109.5° (C) 105.5° (D) 108.5°
11. In which of the following pairs, the two species are isostructural:
 (A) BrO_3^- and XeO_3
 (B) SF_4 and XeF_4
 (C) SO_3^{2-} and NO_3^-
 (D) BF_3 and NF_3
12. Predict the correct order:
 (where bp is bonded pair and lp is lone pair of electrons)
 (A) bp - bp > lp - bp > lp - bp
 (B) lp - bp > bp - bp > lp - lp
 (C) lp - lp > lp - bp > bp - bp
 (D) lp - lp > bp - bp > lp - bp
13. The mathematical expression of first law of thermodynamics is:
 (A) $\Delta U = q$ (B) $\Delta U = W$
 (C) $\Delta U = q + W$ (D) $\Delta U = W_{ad}$
14. $\Delta_f U^\ominus$ of formation of $CH_4(g)$ at certain temperature is -393 kJ mol^{-1} . The value of $\Delta_f H^\ominus$ is:
 (A) Zero (B) $< \Delta_f U^\ominus$
 (C) (D) Equal to $\Delta_f U^\ominus$
 $> \Delta_f U^\ominus$
15. Internal energy does not include:
 (A) Vibrational energy (B) Rotational energy
 (C) Energy arising by gravitational pull (D) Nuclear energy
16. The sign of ΔG for a spontaneous and non-spontaneous process respectively are:
 (A) Positive and negative. (B) Negative and positive.
 (C) Zero and positive. (D) Positive and zero.
17. You are given a lump of an unknown metal. You perform an experiment and figure out the specific heat is close to $0.89 \text{ J/g } ^\circ\text{C}$. What could be the identity of the metal?
 (A) Gold (B) Silver
 (C) Aluminum (D) None of the above

18. Which of the following is not an example of redox reaction?

- (A) $\text{CuO} + \text{H}_2 \longrightarrow \text{Cu} + \text{H}_2\text{O}$
(B) $\text{Fe}_2\text{O}_3 + 3\text{CO} \longrightarrow 2\text{Fe} + 3\text{CO}_2$
(C) $2\text{K} + \text{F}_2 \longrightarrow 2\text{KF}$
(D) $\text{BaCl}_2 + \text{H}_2\text{SO}_4 \longrightarrow \text{BaSO}_4 + 2\text{HCl}$

19. Identify the correct statement (s) in relation to the following reaction:



- (A) Zinc is acting as an oxidant. (B) Chlorine is acting as a reductant.
(C) Hydrogen ion is acting as an oxidant. (D) Zinc is acting as a reductant.

20. What is the oxidation number of lithium in LiCl ?

- (A) +3 (B) -1 (C) +1 (D) 0

21. What is the oxidation number of O in a diatomic molecule (O_2)?

- (A) +2 (B) ± 2 (C) +8 (D) 0

22. What is the oxidation state of Mn in the compound K_2MnO_4 ?

- (A) 3 (B) 4 (C) 5 (D) 6

23. Which of the following metal displacement reaction will not take place and why?

- (A) $\text{Cu} + \text{Mg}^{2+} \longrightarrow$ (B) $\text{Mg} + \text{Cu}^{2+} \longrightarrow$
(C) $\text{Pb} + \text{Ag}^+ \longrightarrow$ (D) $\text{Zn} + \text{Cu}^{2+} \longrightarrow$

*** a statement of Assertion (A) is followed by a statement of Reason (R). [2]**
Choose the correct option.

24. **Note:** In the following questions a statement of Assertion (A) followed by a statement of Reason (R) is given. Choose the correct option out of the choices given below each question.

Assertion (A): Though the central atom of both NH_3 and H_2O molecules are sp^3 hybridised, yet $\text{H}-\text{N}-\text{H}$ bond angle is greater than that of $\text{H}-\text{O}-\text{H}$. **Reason (R):** This is because nitrogen atom has one lone pair and oxygen atom has two lone pairs.

- i. A and R both are correct, and R is the correct explanation of A.
ii. A and R both are correct, but R is not the correct explanation of A.
iii. A is true but R is false.
iv. A and R both are false.

25. Direction in the following Questions, the Assertion and Reason have been put forward. Read the statements carefully and choose the correct alternative from the following:

- a. Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
- b. Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
- c. Assertion is true but Reason is false.
- d. Both Assertion and Reason are false.

Assertion: Ionic compounds usually have high melting and boiling points.

Reason: A large amount of energy is needed to overcome the strong interionic electrostatic attractive forces.
